

Assignment: Text Mining and Sentiment Analysis

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1 Introduction

When buying something online, the first thing most people do is check the reviews. Platforms like Amazon collect millions of customer evaluations in which they express their opinions, experiences, and assessments of products. However, for companies, manually reviewing all these comments is impossible. To overcome this obstacle, Text Mining and Sentiment Analysis have become increasingly popular in recent years, allowing researchers to automatically analyze thousands of reviews using computational methods and tools such as R, in order to assess consumer satisfaction.

For this assignment, we chose to analyze the sodastream Fizz & Go Blueberry, an insulated bottle sold on Amazon.co.uk. As a lifestyle product designed for daily use and sustainability, it represents an interesting case for understanding consumer judgment for experiential products.

Accordingly, this study addresses the following research question: what is the overall sentiment expressed by Amazon customers toward this product, and which attributes, such as quality, durability, and thermal performance are most frequently linked to positive or negative evaluations?

2 Data

The dataset used for this case study was collected directly from [Amazon.co.uk](https://www.amazon.co.uk) by scraping the product page of the [sodastream Fizz & Go Blueberry](https://www.amazon.co.uk/sodastream-fizz-go-blueberry).

Since Amazon employs dynamic web structures and security walls, data collection was performed using the `RSelenium` package in R to handle the login phase and navigate through the review pages, combined with `rvest` to parse the HTML content.

A total of 100 reviews were initially extracted. For each review, we collected three primary variables:

- **Review title:** short headline summarizing the user's opinion.
- **Review text:** full textual comment containing the detailed customer experience.
- **Star rating:** explicit quantitative evaluation left by the customer (ranging from 1 to 5 stars).

2.1 Data cleaning and preprocessing

Before starting the analysis, the raw data required a screening process to ensure data quality and compatibility with English text mining dictionaries:

1. **Language filtering:** since the product is sold on a global marketplace, some international reviews were present. Using the `clld2` library, we analyzed the text and titles to detect the language. We filtered the dataset to keep only reviews written in English (or cases where the language could not be determined but the text was relevant).
2. **Score extraction:** the original star rating was scraped as a text string (e.g., "4.0 out of 5 stars"). Using regular expressions via the `stringr` package (`\\d+\\.\\d+`), we extracted the numeric value into a new variable called score.

2.2 Descriptive Statistics

The final dataset shows a general overview of customer satisfaction. Based on the summary statistics calculated in R, the product has an average rating of 4.18 stars.

As illustrated in Figure 1, the distribution of the scores is highly skewed towards positive evaluations. In fact, 5-star reviews account for **53%** of the entire dataset, followed by 4-star reviews at **30%**. Conversely, negative feedback (1 and 2 stars) represents a minor portion of the sample (**11%**). This distribution indicates a generally high level of customer satisfaction, which will be further analyzed through textual sentiment analysis to understand the specific drivers behind these scores.

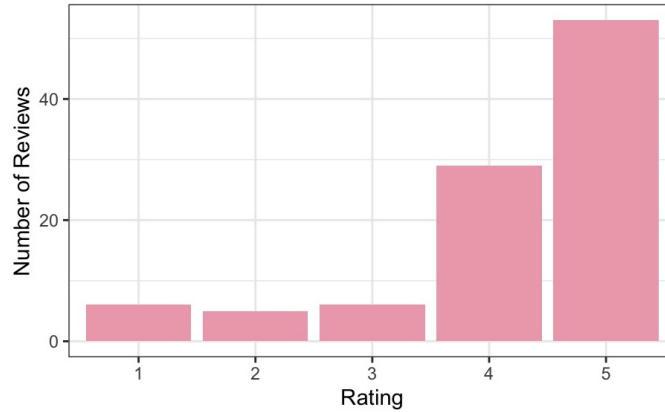


Figure 1: Amazon Review's star ratings.

3 Results and Discussion

3.1 Descriptive Analysis and Text Exploration

To understand the qualitative feedback, we analyzed word frequencies. The removal of standard stop-words was a critical step here; without it, the top words would be meaningless syntax fillers like *the* or *and*.

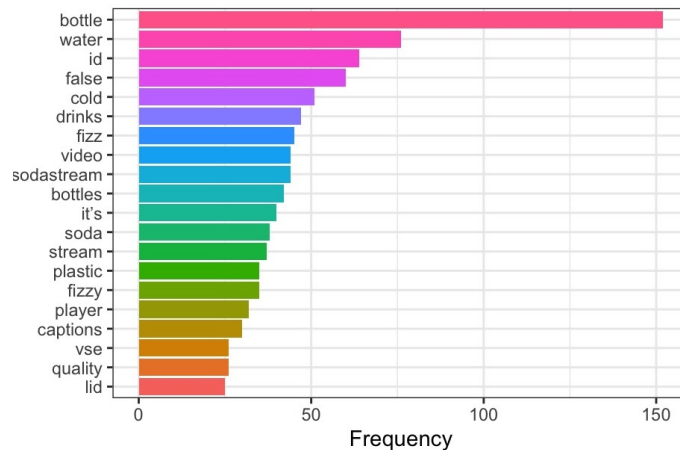


Figure 2: Top 20 most frequent words.

As shown in Figure 2, the most frequent words directly reflect the product features of the sodastream Fizz& Go Blueberry. Words like *bottle*, *water*, *cold*, *drinks*, and *fizz* dominate the text, proving that users heavily discuss its main functions: keeping carbonated water cold.

However, the frequency plot also reveals some noise. Terms like *video*, *player*, *captions*, and *false*

appear in the top 20. This is an interesting text mining byproduct: because we scraped raw HTML review pages from Amazon, the code inadvertently captured text elements from the embedded Amazon video review player rather than just user commentary.

3.2 Dictionary-Based Sentiment Analysis

In the next step, we performed an unsupervised sentiment analysis using the Bing lexicon. We compared the classic tidy approach (which counts single words independently) against the syntax-aware `udpipe` approach.

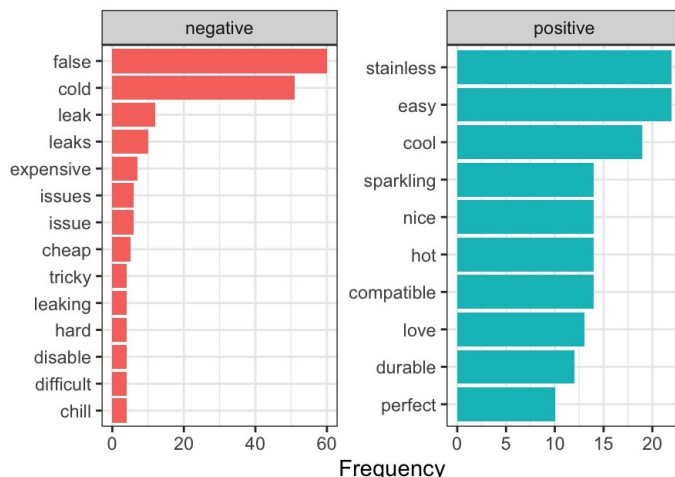


Figure 3: Most frequent positive and negative words.

As shown in Figure 3, one of the most significant limitations of generic sentiment dictionaries is **contextual misclassification**.

For instance, the Bing lexicon categorizes the words *cold* and *chill* as negative. In a general context, this may hold true as in *cold weather* or *a cold reception*. However, it becomes highly problematic when analyzing reviews of an insulated stainless steel bottle, whose core selling point is precisely its ability to keep beverages cold.

As a result, the classic `tidytext` approach artificially lowers the sentiment score of highly positive reviews that praise the product’s thermal performance.

A further distortion arises from the word *false*, which appears as the top negative word, not because it appears in critical reviews, but because it was extracted from the video player interface layout (*false/true* values in HTML), rather than from actual negative content. This highlights an additional weakness of purely lexicon-based methods: their inability to distinguish between natural language and technical or structural text.

The Figure also reveals which words drive the Bing sentiment score. On the positive side, terms such as *stainless*, *easy*, *cool*, and *compatible* match with the nature of the product’s attributes, suggesting that the lexicon captures genuine positive feedback in these cases.

On the negative side, words like *leaks* and *expensive* capture customer concerns, confirming that the dictionary is more reliable when domain-specific meanings align with their natural usage.

The `udpipe` context approach helps fix some of these flat dictionary errors by looking at words in sequence. By incorporating valence shifters (such as negators *not* or *no*), UDPipe correctly interprets a phrase like *not easy* as negative, whereas the TidyText approach, operating word by word, would isolate *easy* and mistakenly count it as a positive signal.

3.3 Supervised Machine Learning

To assess if a machine learning approach could handle better the limitations identified above, we built a supervised **Naïve Bayes classifier** using the **quanteda** package. Given that 3-star reviews were under-represented in the dataset (6 out of 99), they were removed in order to construct a clean binary model:

- **Negative class:** 1 and 2-star reviews.
- **Positive class:** 4 and 5-star reviews.

Word stemming was applied to reduce grammatical variations (e.g., mapping both *leaking* and *leaks* to the root *leak*). The dataset was then partitioned into a 70/30 training and test split using a fixed random seed.

3.4 Methodological Performance Evaluation

To determine which text mining approach generates the most reliable results, we compared the accuracy scores of all four implementations on the same test dataset:

- **Tidy Bing Accuracy:** 0.786
- **UDPipe Basic Accuracy:** 0.821
- **UDPipe Context Accuracy:** 0.857
- **Naive Bayes Accuracy:** 0.857

The comparison reveals that both **UDPipe Context** and **Naive Bayes** achieved the highest overall accuracy. Dictionary-based approaches are useful because they do not require training data. However, they suffer from the general nature of their vocabulary. As mentioned previously, the word *cold* is classified as negative.

The UDPipe method improves accuracy because it catches grammatical rules and negations. Nevertheless, the Naïve Bayes classifier demonstrates equally strong performance to UDPipe Context, as it does not rely on fixed external dictionaries. Instead, it learns word-level probabilities directly from the target dataset. In this case, Amazon product reviews.

4 Conclusions

This case study successfully analyzed customer perceptions of the sodastream Fizz & Go Blueberry on [Amazon.co.uk](https://www.amazon.co.uk) reviews. Overall consumer sentiment emerges as strongly positive: customers highly value the product’s premium stainless steel design and its effectiveness in maintaining carbonated beverages cold. Where negative feedback does appear, it involves specific complaints about structural issues such as leakage and perceived price.

From a methodological perspective, the project highlights the importance of selecting the right text mining framework. Flat dictionary-based methods such as **tidytext** are fast to implement but remain blind to linguistic context, making them prone to systematic errors when domain-specific terminology or negation structures are involved.

Adding syntactic analysis through **udpipe**, together with valence shifters, significantly reduces these context related errors.

Ultimately, the supervised Naïve Bayes classifier proves to be the most robust solution for automated sentiment analysis, as it adapts dynamically to the real vocabulary used by customers.

For companies like SodaStream, setting up these automated text mining pipelines offers concrete

strategic value: it enables real time monitoring of customer satisfaction, support the early detection of product defects (like leakage), and allows marketing efforts to be aligned more precisely with the features that resonate most strongly with consumers.

References

- [1] Amazon.co.uk. sodastream fizz & go blueberry. <https://www.amazon.co.uk/dp/B0GJF22LYF>, 2026.